

Data Enrichment Documentation

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This documentation outlines the automated data enrichment process for the fully annotated Economies of Space dataset.¹ Please note that this document is neither an annotation manual nor a discussion of the historical sources. For annotation guidelines, please refer to the documentation of the manually annotated dataset,² as this dataset follows the same principles. For information regarding historical sources, please consult the Staatsarchiv entry about the *Historisches Grundbuch der Stadt Basel*.³

Dataset

This dataset comprises 75,273 documents from the *Historisches Grundbuch der Stadt Basel* 1 that meet our project criteria: they are dated between 1400 and 1700 and were identified as being written in German or predominantly in German. Please refer to our paper about the manually annotated dataset for details regarding the HTR process.⁴

Training Data

Several of these steps rely on custom models trained using Flair. Unless otherwise specified, these models were trained on the manually annotated dataset.⁵ Where examples for LLM-supported processes were required, they were also sourced from that dataset.

Span Annotation

We applied minimal preprocessing to the digitized dataset by separating punctuation into distinct tokens. To annotate nested entities, we employed a recursive annotation strategy within the Flair framework, processing from larger to smaller spans.⁶ The model is available on Huggingface.⁷ Each span is categorized with an *element* and *class* attribute. The provided confidence value is derived from the Flair model.

```
<span id="3" start="27" end="30" element="reference" class="per"
  confidence="1.0" text="Henman Kurtzin den Rebman"
  numerus="sgl" specificity="spc" subclass="">
```

Flair document classifiers were trained and applied to determine the *numerus*, *specificity* and *subclass* of each span. The classifier showed an accuracy of 93.11% when predicting the

¹ <https://github.com/DHBern/economies-of-space-automatically-annotated-data>

² <https://doi.org/10.5334/johd.387>

³ <https://dls.staatsarchiv.bs.ch/records/1016781>

⁴ <https://doi.org/10.5334/johd.387>

⁵ <https://doi.org/10.5334/johd.387>

⁶ <https://zenodo.org/records/11500543>

⁷ <https://huggingface.co/dh-unibe/hgb-ner-v2>

three criteria at the same time. The most common error was misclassifying *grp* references for *sgl* references.

We did not implement corrections for unusual or forbidden nesting, with the exception of addressing multiple *head* elements within references. If multiple *head* spans were detected, we compared their confidence scores and assigned the *other* element attribute to all but the span with the highest confidence.

Coreference Annotation

Coreference annotation was not an initial project focus and was limited to specific subsets of spans.

For pronoun references, we trained a sequence tagging model that used the highlighted pronoun and pretagged text to identify potential link candidates. Evaluated on our testset, the model was able to find 61.95% of all coref-targets, and achieved a precision of 82.35%. To improve recall for simpler constructions, we implemented a rule-based algorithm linking pronouns nested inside *fam* appositions to the previously mentioned reference (e.g., resolving “Hans und seine Frau Anna”).

For organizations, we first applied the normalization step (see below) and then identified generic concepts: *Kloster*, *Gotteshaus*, *Kaplanei*, *Konvent*, *Zunft*, *Kirchen*, *Pfleger*, *Schaffnerin*, *Kapelle*. We then used a rule-based algorithm to build coreferences by matching these generic terms with preceding proper name mentions of organizations.

Finally, we utilized a rule-based algorithm to identify references containing explicit markers of repeated entity mentions, such as “vorgenannt”, and attempted to match them with previously identified references using similarity measures.

Certain documents were reviewed by GPT-5.5 (see next section) and contain coreferences generated by the model.

Despite these steps, the precision and recall of coreferences remain suboptimal, and we advise users to exercise caution when utilizing this portion of the dataset.

Event and Relation Annotation

For all events and states, we attempted to determine *tense*, *polarity*, and *modality* using a classifier similar to the one used for the minor classes in our span annotation. However, due to an insufficient number of samples deviating from the default, these models were not effective enough. Consequently, we set these attributes to *unk* to signify missing information.

Nested Events

Nested events and states were generated via a postprocessing script based on the structure of the nested spans. If an event or state resulted from an apposition, the trigger is defined as

the head of that apposition. The *ref* attribute of the event Group points to the ID of the span representing the intermediary layer between the event participants.

```
<eventGroup event_id="1" class="family" start="5" end="11" ref="8" type="state" tense="unk" polarity="unk"
  modality="unk">
  <trigger start="11" end="11" text="ewirtin" ref="10" />
  <event event_id="1.0">
    <role role="family-a" ref="1" text="Seczin Settknen wil . Contzen Bettken des Karrers ewirtin" />
    <role role="family-b" ref="9" text="Contzen Bettken des Karrers" />
  </event>
</eventGroup>
```

Trigger-based Events

Wherever the span annotation model identified a *trigger* element, we employed our role annotation model to identify the participants. This model followed the sequence tagging strategy described in our 2024 paper to assign roles to candidates on the same annotation layer.⁸ The weighted F1-Score is 86.19%, but performance varies strongly depending on event frequency as well as complexity. We thus advise users to evaluate important events and their roles themselves, and if need be, to reannotate or fix them with a different strategy.

```
<eventGroup event_id="0" class="property-purchase" start="2" end="68" ref="" type="event" tense="unk" polarity="unk" modality="unk">
  <trigger ref="0" text="kaufen"/>
  <event event_id="0.0">
    <role role="seller" confidence="1.0" ref="1" text="Seczin Settknen wil . Contzen Bettken des Karrers ewirtin"/>
    <role role="buyer" confidence="0.85" ref="2" text="Elsin ir ehl tochter Heuslis Hugon von Muttentz des Karrers ebl wirtin"/>
    <role role="buyer" confidence="1.0" ref="3" text="Henman Kurtzin den Rebman"/>
    <role role="property" confidence="1.0" ref="5" text="eins hofstettlis u eins gertlis derhinder gelegen , ze ussrest in der vorstatt
    <role role="price" confidence="1.0" ref="6" text="2 lb ."/>
  </event>
</eventGroup>
```

The confidence values are based on the output by that model.

Role-based Events

In some cases, specifically for due-obligation relations and due-payment events, neither nesting nor a trigger could be used to assign roles. To cover these cases, a specialized model using the *money* values as triggers instead was applied to assign roles. This featured an additional advantage that it could detect different events inside the event group (e.g. if multiple beneficiaries received different interest values). We leveraged this as well for previously annotated relations and events to create subevents where needed. This model achieved an 86.95% weighted F1-Score. For the most notable error, the model would confuse *property* and *payer* roles when properties were named in rent roles. This error likely stems from the way these transactions were phrased, as the property would be mentioned as acting entity (“das Haus an der Rebgassen zinst 4 lb”).

Rule-based Events

We applied a rule-based algorithm to create *representation*-relationships. Our algorithm fuzzy-searched for certain markers such as *Vogt*, *Gewalthaber* and *Schaffner* between two entity references. It also modified the roles of other events where these entities were marked as actors and modified them to highlight *actual* actors and *representing* actors.

⁸ <https://ceur-ws.org/Vol-3834/paper79.pdf>

```
<role role="seller" ref="2" text="Gwalthaber Frau Ennelin siner gemaheln" inferred="True"/>
<role role="repr_seller" ref="4" text="Hugli Lutzoll"/>
```

For rent rolls, which would feature a large number of dates and sometimes multiple payers, we applied a rule-based algorithm to split the event into subevents so the information *who paid the rent in which year* would be easier to extract.

```
<event event_id="0.0">
  <role role="date" confidence="0.87" ref="0" text="1571"/>
  <role role="date-missing" confidence="0.99" ref="1" text="1572"/>
  <role role="date-missing" confidence="0.95" ref="2" text="76"/>
  <role role="payer" confidence="0.96" ref="3" text="Michel Krug"/>
</event>
<event event_id="0.1">
  <role role="date" confidence="1.0" ref="4" text="1577"/>
  <role role="payer" confidence="1.0" ref="5" text="Niclaus Hon"/>
  <role role="due-date" confidence="1.0" ref="7" text="uf Pfingsten"/>
  <role role="interest" confidence="1.0" ref="8" text="1 lb ."/>
</event>
<event event_id="0.2">
  <role role="payer" confidence="0.99" ref="9" text="Hauss Joger , der Kürssner"/>
  <role role="date-missing" confidence="1.0" ref="10" text="1578"/>
  <role role="date" confidence="1.0" ref="11" text="1579"/>
  <role role="date" confidence="1.0" ref="12" text="1580"/>
  <role role="date" confidence="1.0" ref="13" text="1581"/>
  <role role="date" confidence="1.0" ref="14" text="1582"/>
  <role role="date-missing" confidence="1.0" ref="15" text="1583"/>
  <role role="date" confidence="1.0" ref="16" text="1584"/>
</event>
```

GPT-5.5 Corrections

We applied GPT-5.5 to all documents that featured an event trigger on document level classified as *other*. The goal was to define the event class correctly and fix roles and coreferences where needed. No changes to the spans were allowed. All documents that were seen by GPT-5.5 are marked as such in their metadata. Roles modified by GPT say so in their *confidence* attribute.

```
<role role="date" confidence="0.64" ref="0" text="uff Dovrstag nach dem Snentag Cantate"/>
<role role="property" confidence="1.0" ref="4" text="dem Hus Hofstatt genant der Imber"/>
<role role="bidder" ref="2" confidence="gpt-5.5" text="Hanns Jry der junge"/>
<role role="other" ref="3" confidence="gpt-5.5" text="Hanns Howenstein"/>
<role role="bid" ref="6" confidence="gpt-5.5" text="1 lb 4sh ."/>
```

Normalization

Please note that normalization was only applied selectively for some of the analyses in our project. The normalization work that was done manually is published in the same repository as the XML dataset.

Occupations

We combined the similarity-matching approach by Goldberg & Moeller⁹ with a Flair document classifier to normalize occupation variants to standard German. A large number of variants had been normalized by hand previously and was used as training data for the classifier. Where applicable, we linked normalizations to their concepts in the database of *Ontologie historischer, deutschsprachiger Berufs- und Amtsbezeichnungen*. We evaluated a sample of 3036 automatically annotated occupation mentions (not distinct variants!) and found that our normalization was correct in 2659 cases. 3 normalizations were wrong. For the other occupations, the pipeline had classified them as *unclear*. We judged that 127 of those could've been resolved to a proper normalization, the other cases were either not occupations, or not readable enough due to HTR errors.

```
<span id="8" start="7" end="7" element="head" class="occ"
  confidence="1.0" text="Spittalmeister" norm="spitalmeister"
  OhdAB="https://database.factgrid.de/wiki/Item:Q693199" />
```

Organizations

For organizations, we applied a Flair document classifier based on manual training data. In a later step, we tried to increase our recall by prompting GPT-4o to normalize some of the organizations that the Flair classifier could not normalize reliably. We evaluated 1013 organization normalizations by hand. 796 were correctly normalized. In 5 cases, the assigned normalizations were wrong. The other cases were marked *unclear* by the system, but we judged 77 of those could've been normalized.

```
<span id="22" start="44" end="47" element="head" class="nam"
  confidence="0.84" text="Pfleger des grossen Allmosen"
  norm="Städtisches Almosen" />
```

Property Types

We employed GPT-4o to normalize *type* heads of *fac* references. The correct normalization was applied in 93.85% of 488 cases, 2.87% were marked *unclear*, 1.84% were actually proper names, and 1.43% were wrong.

```
<span id="12" start="16" end="16" element="head" class="type"
  confidence="1.0" text="Hüsern" norm="Häuser" />
```

Monetary Values

We employed GPT-4o to normalize *money* values. We separately prompted normalization of written-out or more complicated numerical values and normalization of currencies. We provided the model with a target set of currencies appearing in the dataset. We evaluated 1856 currency mentions (not distinct ones!) and found that 95.51% were normalized

⁹ https://doi.org/10.17175/2022_002

correctly, while 4.04% had been marked as *unclear*. No wrong normalizations had been assigned.

```
<span id="4" start="54" end="59" element="value" class="money"
  confidence="0.99" text="42 lb . 10 sh ."
  norm="42.0 Pfund|10.0 Schilling" />
```